Chapter 9: Plant Physiology

Solved Short Questions (30)

1. Define mineral nutrition in plants.

The absorption of mineral elements (nutrients) from the soil, which are essential for the growth and life processes of plants, is called mineral nutrition.

2. Define macronutrients and micronutrients with examples.

Macronutrients are minerals required by plants in large quantities, e.g. nitrogen, phosphorus, potassium, calcium and magnesium. Micronutrients are required in small quantities, e.g. iron, zinc, copper, boron and manganese.

3. State the roles of nitrogen and magnesium in plants.

Nitrogen is a component of proteins, chlorophyll and enzymes. Magnesium is a component of chlorophyll and activates many enzymes in plants.

4. Differentiate between autotrophic and heterotrophic organisms.

Autotrophic organisms prepare their own food from water, carbon dioxide and minerals, e.g. all green plants. Heterotrophic organisms obtain food from other organisms, e.g. animals, fungi and most bacteria.

5. Why is transport necessary in plants?

Roots absorb water and minerals from soil while leaves prepare food. Transport carries water and minerals upward to leaves and food from leaves to all other parts of the plant.

6. Describe the structure of a root hair.

A root hair is a thin, elongated extension of a root epidermal cell. Its thin wall and large surface area help in absorbing water and mineral salts from the soil.

7. How do roots absorb water from the soil?

Roots absorb water by passive absorption: water moves from the soil (higher concentration) into root hairs (lower concentration) by osmosis, without using energy.

8. How do roots absorb mineral salts from the soil?

Mineral salts are absorbed by active absorption: root cells use energy (ATP) to take up mineral ions from the soil even against their concentration gradient.

9. Define transpiration and write its types.

The loss of water in the form of vapours from the plant surface is called transpiration. Its types are: stomatal transpiration (through stomata), cuticular transpiration (through cuticle) and lenticular transpiration (through lenticels).

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10. How does stomatal transpiration take place?

Water moves from xylem into the moist cell walls of mesophyll cells, evaporates into leaf air spaces, and then diffuses out through open stomata into the outside air.

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11. Describe the opening and closing of stomata.

When guard cells absorb water they become turgid and the stoma opens. When they lose water they become flaccid and the stoma closes. Potassium ions regulate this movement of water.

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12. How is transpirational pull important in plants?

Transpirational pull is the upward suction force created by continuous water loss from leaves. It pulls the water column upward through xylem from roots to the top of tall plants.

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13. Is transpiration a harmful phenomenon? Give its importance.

Although water is lost, transpiration is not harmful. It creates the transpirational pull for water transport, cools the plant, and helps in absorption and distribution of minerals.

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14. How do temperature, wind and humidity affect the rate of transpiration?

High temperature increases evaporation, so transpiration rises. Wind removes water vapours from leaf surface, increasing the rate. High humidity in air reduces the rate of transpiration.

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15. Differentiate between xylem and phloem.

Xylem transports water and mineral salts from roots to leaves (upward). Phloem transports prepared food (sugars) from leaves to all other parts of the plant (both directions).

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16. Differentiate between transpiration and guttation.

Transpiration is the loss of water as vapours through stomata, cuticle or lenticels. Guttation is the loss of water as liquid drops through special openings (hydathodes) at leaf margins, usually at night.

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17. What are lenticels? What is lenticular transpiration?

Lenticels are special pores in the bark of woody stems that allow gaseous exchange. Loss of water vapours through lenticels is called lenticular transpiration.

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18. Describe the mechanism of transport of water and salts in plants.

Water and salts absorbed by roots move into xylem. Transpirational pull from leaves draws this water column upward through xylem vessels to all aerial parts of the plant.

19. Explain the Pressure Flow Mechanism of food translocation.

In the Pressure Flow Mechanism, sugars made in leaves are loaded into phloem, water follows by osmosis creating high pressure, which pushes the sugary sap toward roots and other parts where pressure is low.

20. Describe the process of gaseous exchange in plants.

In leaves and young stems, air moves in and out through stomata. In the day, CO_2 enters for photosynthesis and O_2 is released; at night, O_2 is taken in and CO_2 released during respiration. In woody stems, exchange occurs through lenticels.

21. How do plants excrete their waste products?

Plants store wastes in old leaves which fall off, in bark which is shed, in fruits, or excrete them as resins, gums and latex. Excess water and salts are removed through guttation and transpiration.

22. How do rubber and keekar plants excrete their wastes?

Rubber plants excrete waste as latex, while keekar (acacia) plants excrete waste as gum through their stems.

23. Define hydrophytes, xerophytes and halophytes.

Hydrophytes are plants that grow in water. Xerophytes grow in dry/desert conditions. Halophytes grow in salty soils.

24. What are osmotic adjustments in plants?

Osmotic adjustments are the changes plants make in their cell sap concentration to survive in different environments — e.g. xerophytes store water in succulent organs, halophytes accumulate salts to absorb water from salty soil.

25. State the role of potassium in plants.

Potassium regulates the opening and closing of stomata by controlling water movement in guard cells.

26. Why is nitrogen called an essential macronutrient?

Nitrogen is needed in large quantity because it is a component of proteins, chlorophyll, enzymes and nucleic acids — all essential for growth.

27. Name the two pathways of water movement inside the root.

Inside the root, water and salts move toward the xylem by two pathways: (i) through the cells, and (ii) through cell walls and intercellular spaces.

28. What is the role of root hairs in absorption?

Root hairs greatly increase the surface area of the root, which helps in absorbing maximum water and mineral salts from the soil.

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29. Why do plants transpire more on a windy day than on a humid day?

Wind continuously removes water vapours from around the leaves, maintaining a steep diffusion gradient, so transpiration increases. On humid days the air already holds much vapour, so the rate falls.

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30. What is the importance of magnesium for chlorophyll?

Magnesium is the central atom of the chlorophyll molecule. Without it, chlorophyll cannot form and photosynthesis stops.
